

int mat[5][6]

rows

columns

	0	1	2	3	4	5
0						
1						
2						
3						
4						

mat[1][3]

mat[3][3]

int mat[N][M]

TL:- mat[0][0]

TR:- mat[0][M-1]

BL:- mat[N-1][0]

BR:- mat[N-1][M-1]

iterate on a row in matrix
row index is same, and col $\rightarrow [0, M-1]$

iterate on a col in matrix.
col. index is same, and row $\rightarrow [0, N-1]$

	0	1	2	...	M-1
0		0,1			
1		1,1			
2	2,0	2,1	2,2	-	2,M-1
...		:			
N-1		N-1,1			

Ques 1 Given $mat[N][M]$, print row-wise sum.

e.g. $mat[3][4]$

	0	1	2	3
0	3	8	9	2
1	1	2	3	6
2	4	10	11	17

Sum:-

22

12

42

// Pseudo code:-

```
for (i=0; i < N; i++) {
    int s=0;
    for (j=0; j < M; j++) {
        s = s + mat[i][j];
    }
    print(s);
}
```

T.C = $O(N \times M)$

S.C = $O(1)$

Ques 2 Given $mat[N][M]$, find max column sum.

	0	1	2	3
0	3	8	9	2
1	1	2	3	6
2	4	10	11	8
	8	20	23	16

Ans

Standard:-

i → for rows

j → for columns.

→ Initialize with 0 or $mat[0][0]$

ans = INT_MIN

```
for (j=0; j < M; j++) {
    sum=0;
    for (i=0; i < N; i++) {
        sum = sum + mat[i][j];
    }
    ans = max(ans, sum);
}
return ans;
```

T.C = $O(N \times M)$

S.C = $O(1)$

Ques 3 Given a $\text{mat}[N][N]$, print diagonals.
 ↗ left → right → TL → BR
 ↘ right → left → TR → BL
 square matrix

→ $\text{mat}[4][4]$

	0	1	2	3
0	0,0			0,3
1		1,1	1,2	
2		2,1	2,2	
3	3,0			3,3

① left to right

i and j are same

$i = 0$

while ($i < N$) {

 print(mat[i][i]);

 i++;

}

T.C = $O(N)$

S.C = $O(1)$

② right to left

$i = 0; j = N-1$ ↗ $j \geq 0$

while ($i < N$) {

 print(mat[i][j])

 i++; j--;

}

T.C = $O(N)$

S.C = $O(1)$

i	j
0	3
1	2
2	1
3	0
4	-1

total T.C = $O(2N) = O(N)$

S.C = $O(1)$

Ques 4 Given $mat[N][M]$, print all diagonals going $R \rightarrow L$

$mat[4][6]$

	0	1	2	3	4	5
0						
1						
2						
3						

Observations:-

1) Diagonals are starting from 0th row and last column.

$\downarrow$
M-1th col.

$\langle 0, 5 \rangle$ $\langle 0, 4 \rangle$ $\langle 1, 5 \rangle$
 $\langle 1, 4 \rangle$ $\langle 1, 3 \rangle$ $\langle 2, 4 \rangle$
 $\langle 2, 3 \rangle$ $\langle 2, 2 \rangle$ $\langle 3, 3 \rangle$
 $\langle 3, 2 \rangle$ $\langle 3, 1 \rangle$ $\langle 4, 2 \rangle$
 $\times \langle 4, 1 \rangle$

$\downarrow$
 col - 1
exceeds
 $\downarrow$
 row exceeds
 it is not possible

$mat[3][5]$

	0	1	2	3	4
0	1	2	3	4	5
1	6	7	8	9	10
2	11	12	13	14	15

output

1
 2 6
 3 7 11
 4 8 12
 5 9 13
 10 14
 15

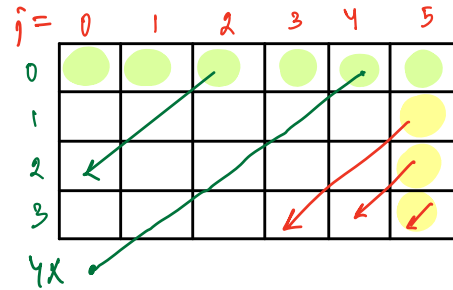
$\rightarrow$ Row and col, neither should exceed.

$\rightarrow$ Both conditions are imp.

// Pseudo Code:-

Step 1:- Print all diagonals starting from 0th row.

```
for(j=0; j<M; j++) {
    x=0; y=j;
    while (x<N & y<=0) {
        print(mat[x][y]);
        x++; y--;
    }
    print("\n");
}
```



Step 2:- Print all diagonals starting from (M-1)th col.

```
for(i=1; i<N; i++) {
    x=i; y=M-1;
    while (x<N & y>=0) {
        print(mat[x][y]);
        x++; y--;
    }
}
```

T.C = $O(N \times M)$
S.C = $O(1)$

i	x	y
1	1	5
	2	4
	3	3
2	1	2
3	2	1
4	3	0

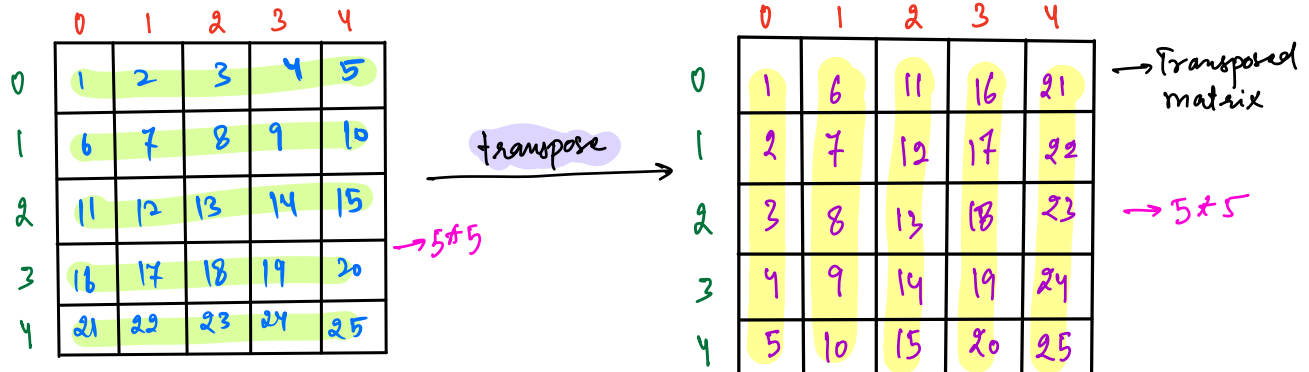
$$\rightarrow (N-1) + (N-2) + (N-3) + \dots + 1$$

$\textcircled{N^2}$

Break $\rightarrow$ 10:42 PM

Ques 5 Given a $\text{mat}[N][N]$, find the transpose in place
 Given input $\text{mat}[1][1]$, should update. S.C = $O(1)$
Square matrix

$\text{mat}[5][5]$



$\text{mat}[3][4]$

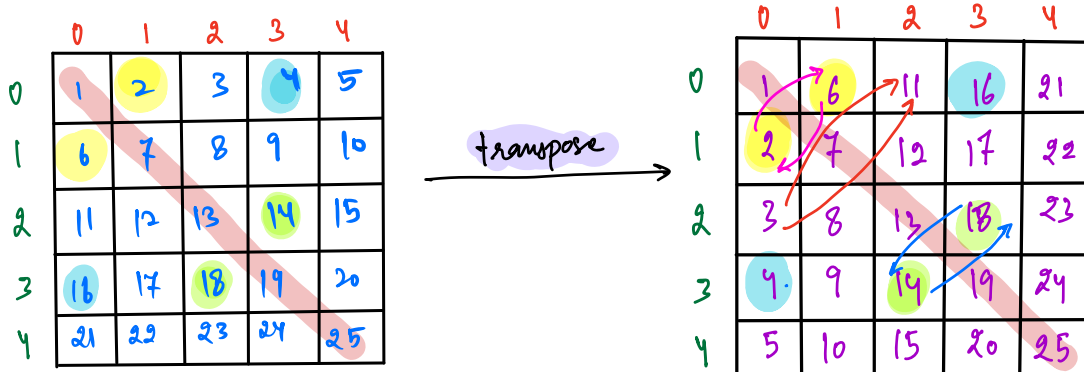
	0	1	2	3
0	1	2	3	4
1	5	6	7	8
2	9	10	11	12

	0	1	2
0	1	5	9
1	2	6	10
2	3	7	11
3	4	8	12

TMat [4][3]

→ The size has changed
 → Cannot transpose a rectangular matrix in place.

Idea?



Observations:-

1) Data does not change for the $(l \rightarrow r)$ diagonal.

2) Swap $mat[0][1]$ with $mat[1][0]$
Swap $mat[3][2]$ with $mat[2][3]$
Swap $mat[0][3]$ with $mat[3][0]$

Swap $mat[i][j]$ with $mat[j][i]$

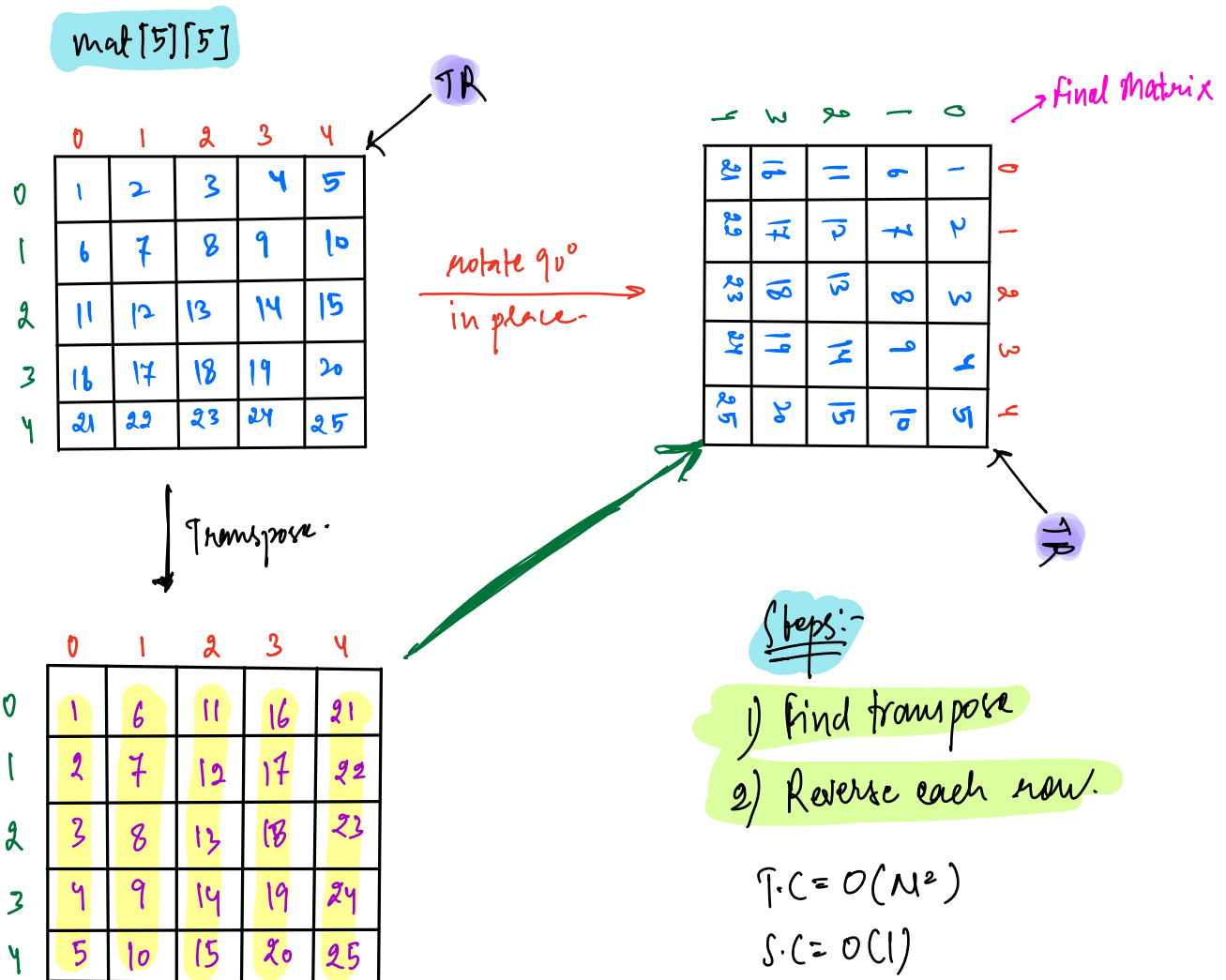
```
for(i=0; i<N; i++){  
    for(j=0; j<M; j++){  
        Swap  $mat[i][j]$  with  $mat[j][i]$   
    }  
}
```

Issue:- We need to do the swapping for only either the lower Δ or the upper Δ .

M.V. $\rightarrow$ iterate on lower Δ and swap $\rightarrow$ transpose matrix

$$T.C = O(N^2)$$

Ques 6 Given a square matrix rotate by 90° in clockwise from TR as reference, in place.



Advance $\rightarrow$ 1 2 D
 &
Advance Questions.

Double:-

	0	1	2	3
0	4	-2	-100	-5
1	-2	0	1	-3
2	-3	0	0	0

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 -1 -2 -99 -8

$$\text{mat}[0][0] = 4$$

$$\underline{\underline{\text{ans} = 1}}$$

$$\max(4, -1) \rightarrow 4$$

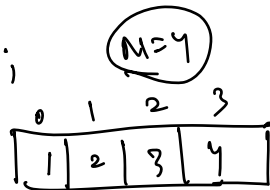
$$\max(4, -2) \rightarrow 4$$

$$\max(4, -99) \rightarrow 4$$

$$\max(4, -8) \rightarrow 4X$$

* int[] res =

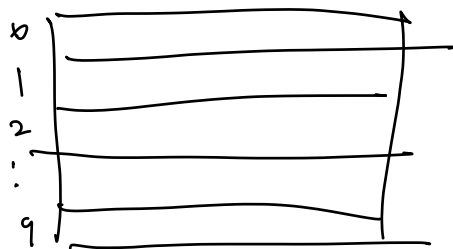
return res;



int[] res.

$$N(N+1)/2.$$

$$4(5)/2 = 10$$



int[N(N+1)/2][N] = new int[7][7];

S, e.